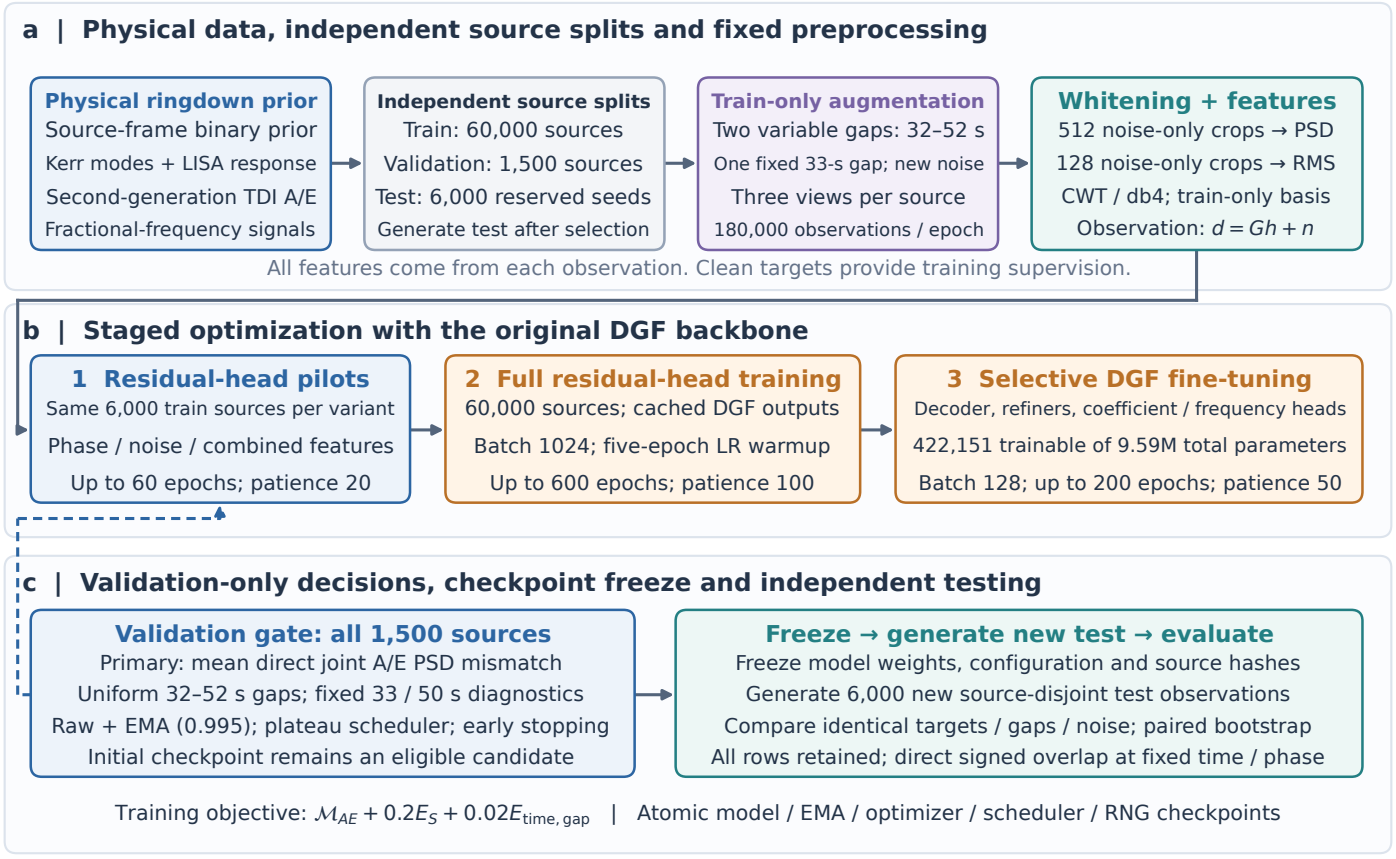
a | Physical data, independent source splits and fixed preprocessing
Physical ringdown prior
Source-frame binary prior
Kerr modes + LISA response
Second-generation TDI A/E
Fractional-frequency signals
Independent source splits
Train: 60,000 sources
Validation: 1,500 sources
Test: 6,000 reserved seeds
Generate test after selection
Train-only augmentation
Two variable gaps: 32–52 s
One fixed 33-s gap; new noise
Three views per source
180,000 observations / epoch
Whitening + features
512 noise-only crops → PSD
128 noise-only crops → RMS
CWT / db4; train-only basis
Observation: $d = Gh + n$
All features come from each observation. Clean targets provide training supervision.
b | Staged optimization with the original DGF backbone
1 Residual-head pilots
Same 6,000 train sources per variant
Phase / noise / combined features
Up to 60 epochs; patience 20
2 Full residual-head training
60,000 sources; cached DGF outputs
Batch 1024; five-epoch LR warmup
Up to 600 epochs; patience 100
3 Selective DGF fine-tuning
Decoder, refiners, coefficient / frequency heads
422,151 trainable of 9.59M total parameters
Batch 128; up to 200 epochs; patience 50
c | Validation-only decisions, checkpoint freeze and independent testing
Validation gate: all 1,500 sources
Primary: mean direct joint A/E PSD mismatch
Uniform 32–52 s gaps; fixed 33 / 50 s diagnostics
Raw + EMA (0.995); plateau scheduler; early stopping
Initial checkpoint remains an eligible candidate
Freeze → generate new test → evaluate
Freeze model weights, configuration and source hashes
Generate 6,000 new source-disjoint test observations
Compare identical targets / gaps / noise; paired bootstrap
All rows retained; direct signed overlap at fixed time / phase
Training objective: $\mathcal{M}_{AE} + 0.2E_S + 0.02E_{\text{time, gap}}$ | Atomic model / EMA / optimizer / scheduler / RNG checkpoints